Loading and Unloading Process Guide

Last updated: 02/08/2026

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Loading inventory is a very common step in most warehouses. It is usually the last point where the product is directly handled by the warehouse associates. It represents the process by which users pick up and place oLPNs and/or pallets onto trailers for transportation to their destination. This process can take many forms depending on the order mix, shipping mode, and physical organization of inventory for transit.

The system offers a robust mobile solution for driving and supporting the loading process. It supports both a non-directed operations-driven model of loading, and a system-driven/directed form of loading. You can initiate loading by accessing a loading task that was created for the oLPN/pallet or by directly scanning the oLPN/pallet in the loading transaction. You can plan shipments ahead of time or assign oLPNs directly to shipments at the time of loading. Loading is not, however, a mandatory step; the system provides a means to invoke loading behind the scenes automatically, or to complete/ship orders without loading them at all.

By contrast, unloading is primarily used as an exception process, since it represents a step backwards in the inventory's path towards its delivery destination. Reasons for invoking unloading are varied, but most typically they revolve around a need to remove the inventory in order to either adjust or cancel an order, or to reorganize inventory on the truck and send the unloaded inventory on another shipment.

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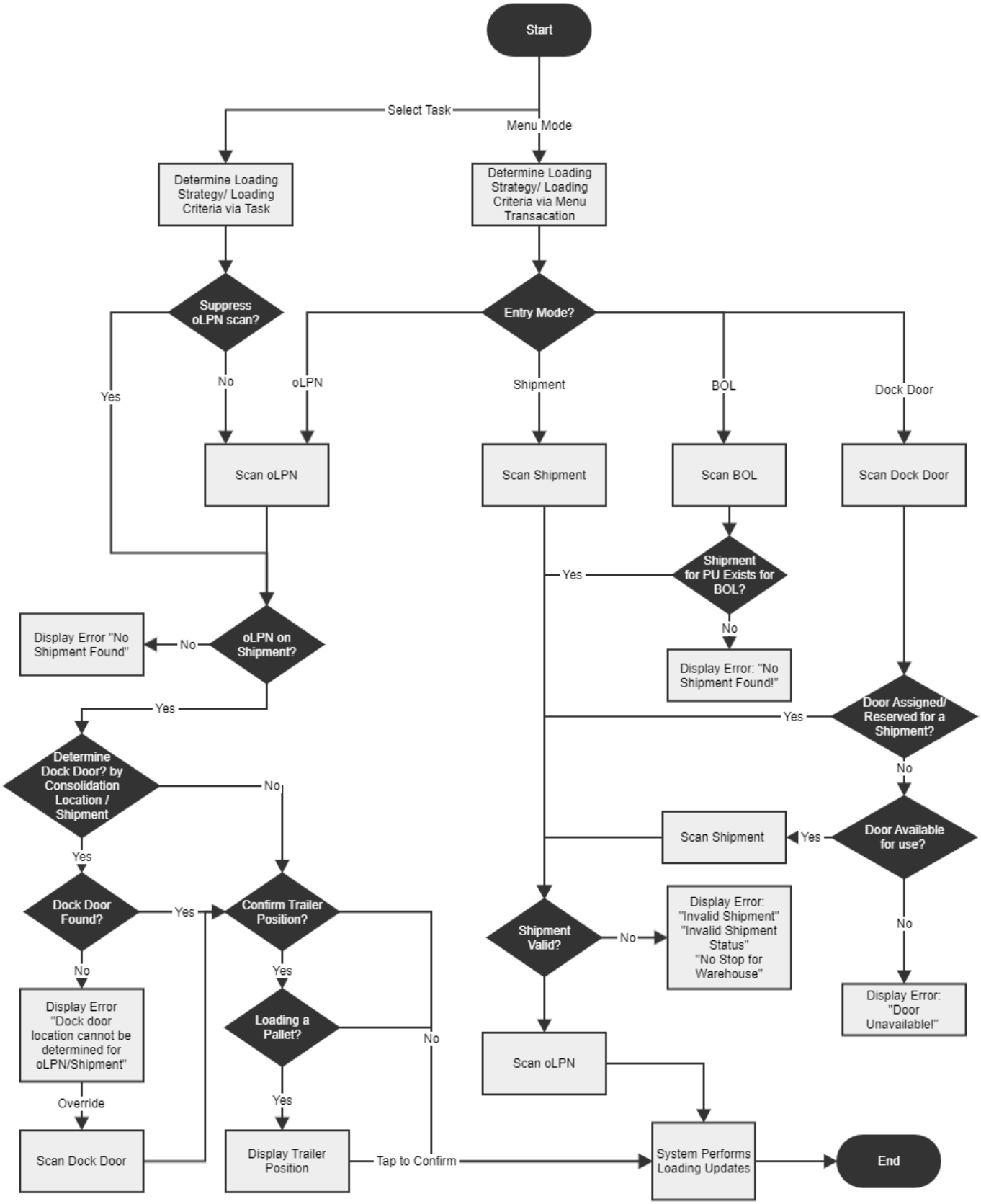
Start
Select Task
Menu Mode
Determine Loading Strategy/ Loading Criteria via Task
Determine Loading Strategy/ Loading Criteria via Menu Transacation
Suppress oLPN scan?
Entry Mode?
Yes
No
oLPN
Shipment
BOL
Dock Door
Scan oLPN
Scan Shipment
Scan BOL
Scan Dock Door
Shipment for PU Exists for BOL?
Yes
No
Display Error: "No Shipment Found!"
Display Error "No Shipment Found"
No
oLPN on Shipment?
Yes
Door Assigned/ Reserved for a Shipment?
Yes
No
Determine Dock Door? by Consolidation Location / Shipment
No
Yes
Scan Shipment
Yes
Door Available for use?
Dock Door Found?
Yes
Confirm Trailer Position?
Shipment Valid?
No
Display Error: "Invalid Shipment" "Invalid Shipment Status" "No Stop for Warehouse"
No
No
Yes
Display Error "Dock door location cannot be determined for oLPN/Shipment"
Loading a Pallet?
No
Display Error: "Door Unavailable!"
Scan oLPN
Override
Yes
Scan Dock Door
Display Trailer Position
Tap to Confirm
System Performs Loading Updates
End

Configuration Summary

Loading

Loading Strategy

The loading strategy defines the overarching business configuration for the loading process. Please refer to the Loading and Unloading Configuration Guide for more details.

Loading Criteria

The loading criteria defines the loading behavior through a set of parameters. The set of parameters determines the screen flow, the validation, and post-processing capabilities. Please refer to the Loading and Unloading Configuration Guide for more details.

Loading Override Criteria

You can override some of the parameters in the loading criteria with another set of parameters by configuring the loading override criteria and linking this to the loading criteria. Please refer to the Loading and Unloading Configuration Guide for more details.

Unloading

Unload oLPN Strategy

The unload oLPN strategy defines the overarching business configuration for the unloading process. Please refer to the Loading and Unloading Configuration Guide for more details.

Unload oLPN Criteria

Unload oLPN criteria defines the unloading behavior through a set of parameters. The set of parameters determines the screen flow, the validation and post-processing capabilities. Please refer to the Loading and Unloading Configuration Guide for more details.

Transaction Configuration

In order for you to use the loading or unloading criteria, you must configure a transaction of type Load or UnloadOlpn in the DC Shipping Transactions screen. This transaction links the Menu Item to a Strategy, for both Loading and Unloading. It dictates what the core set of rules are (Criteria) to be evaluated when the user invokes a given transaction. This is a common configuration approach for all types of user transactions.

Types of Loading

Loading can be divided broadly into two categories based on the nature of the activity - Directed and Non-Directed. Directed loading is a mode in which the user identifies the oLPN, and the system directs them (if possible) to the destination door for that oLPN. Non-Directed loading is a mode in which the user identifies the door, and the system allows them to then scan a series of oLPNs to load at that door. These flavors are described in more detail below.

Directed Loading - "Load oLPN"

Directed loading, also known as the Load oLPN flow, is the mode in which the first prompt for the user is the oLPN. This mode allows a user to select an oLPN and the system directs them to the trailer (typically) for which it is planned. The oLPN must already be assigned to a shipment. The system does not maintain the context of the shipment across multiple oLPNs, since the next oLPN scanned could be for a different shipment. If the system is configured to determine the dock door by shipment, but the shipment is not assigned to a dock door, the system prompts the user to scan a dock door and assign it. After scanning the dock door, the system directs all future oLPNs for this shipment to this door.

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Non-Directed Loading - "Load Trailer"

Non-directed loading, also known as the Load Trailer flow, is the mode in which the user dictates the shipment to load into and subsequently loads oLPNs to it in series.
Depending on how the Loading Criteria for the transaction is configured, the first prompt for the user can be:

- The Door
- The BOL# for the Shipment
- The Shipment ID

This loading mode enables warehouses to staff one or more loading associates at a door and have them load the trailer in a single continuous activity. It reduces the number of scans by only requiring the door scan once up front. The shipment context is maintained until the user exits the transaction - all subsequent oLPNs that are scanned are loaded against the initially-determined shipment. The Load Trailer flow also supports assigning oLPNs to the shipment. In order to accomplish this, you must enable the "Allow Fluid Loading" parameter on the loading criteria.

Loading Functionality

Initiation of Loading

Menu Mode

When the user launches a transaction from the Mobile Menu, the system evaluates the type of transaction; if it is a loading transaction, loading attempts to fetch the loading strategy ID associated to the transaction.

- If no valid strategy is determined, the system displays an error message.
- Otherwise, if it determines a valid strategy, the system evaluates the rules to determine the criteria. This drives the first screen displayed for the user to begin execution.
  - If you have not defined a criteria or rule evaluation does not return a criteria, the system displays the error message "Invalid Criteria definition".
  - Otherwise, if a valid criteria is defined, the system evaluates the "Entry Mode" parameter.
    - If this is set to "oLPN", the first screen prompts the user to scan the oLPN.
    - If this is set to "Shipment," the first screen prompts the user to enter a Shipment ID-- and so forth ("BOL" entry mode prompts the user for BOL number, "Dock Door" prompts the user for Dock Door Barcode or Display Location).

Select/ Request Task Mode

When the user chooses the Select Task functionality (either by requesting next task or providing the task ID or oLPN ID in the enter task mode) from the Main Menu to initiate a task, and a valid loading task is found or provided, this task is passed to the loading function. Loading attempts to fetch the loading strategy ID from the task ID.

- If no valid strategy is determined, the system displays an error message.
- Else if a valid strategy is determined, the system evaluates the rules to determine the criteria. This drives the first screen displayed for the user to begin execution.
  - If you have not defined a criteria or rule evaluation does not return a criteria, the system displays the error message "Invalid Criteria definition".
  - Else, if a valid criteria is defined, the system determines the value for the criteria parameter Suppress oLPN Scan.
    - If set to No (or null), the system prompts the user to confirm the oLPN with a scan.

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- If Yes, the system suppresses the oLPN scan inside of loading and proceeds as if the oLPN were already confirmed, and launches into the remainder of the flow as dictated by the selected Criteria.

Note: When planning and executing a loading task, the transaction must be configured with Entry Mode = oLPN.

Loading task Creation

There are two main ways to create loading tasks.

- Outbound Putaway: As a part of outbound putaway, you can configure the transaction to create loading tasks. The system can be configured to assign the task to the current user immediately and direct the user to load the oLPN, or leave the task for another user to complete. Please refer to the Outbound Putaway Process Guide for more details.
- Shipment UI action: You can also manually trigger the creation of the loading tasks for a shipment by selecting a shipment in the shipment ui and using the "Create Load Task" action.

Directed Loading (Load oLPN) Process

In the directed loading process flow, the sequence of screens starts with the oLPN and ends with the Shipment/Dock Door verification. This process (Container to Door, Container to Door) repeats indefinitely.

Accept oLPN

When entry mode is set to 'oLPN', the first prompt in the Loading transaction (assuming Menu Mode) is for the user to scan/enter an oLPN ID. This initiates a directed loading mode in which the system directs the user to load that oLPN to a shipment to which it is already assigned by Transportation Planning.

When the user identifies an oLPN, the system performs several validations:

- The oLPN must be valid and present in the system. This validation checks that the scanned value is either an oLPN ID or an oLPN Cross-reference ID.
- The oLPN must be in a valid status for loading – at least packed and not already loaded, and not canceled.
- The oLPN must be assigned to a shipment – identified by the Shipment ID on the oLPN itself.
- If stop sequence validation is enabled, then the system validates that there are no other unloaded oLPNs of a higher stop sequence. See below for specific validations for enforcing correct delivery stop sequence loading.
- If stop sequence validation is disabled, then there is an option to enable loading sequence validation. When loading sequence validation is enabled, then the system ensures that the oLPNs for the shipment are being loaded in an ascending loading sequence.

Accept Dock Door

Once the oLPN is validated, if the parameter "Determine Dock Door" is set to ‘Determination of dock door by Consolidation Location' or 'Determine dock door by Shipment’, the system attempts to determine a dock door for the oLPN. If it is found, the system prompts the user to scan the dock door's location to confirm they are loading it at the right door.

- When "Determine Dock Door" is set to "**Determination of dock door by Consolidation Location",** the system checks the oLPN’s current consolidation location and associated dock door. If not found, then the system checks for the oLPN’s destination consolidation location and associated dock door.
  - If the dock door is not found, then the system shows an error message, 'Dock door location cannot be determined for oLPN {olpn_id}'
    - If this error message is configured as a warning and this warning is bypassed, the system allows the user to scan the dock door and proceed for loading.

- When "Determine Dock Door" is set to "**Determine dock door by Shipment**", the system checks for an associated dock door for the oLPN's shipment.
  - If a dock door is not found, then the system shows an error message, 'Dock door location cannot be determined for shipment {shipmentId}'
    - If this error message is configured at a "Warning" level, and you bypass the warning, the system allows the user to scan the dock door and proceed for loading.
  - If the oLPN has a ParentShipmentID populated, this means the shipment the oLPN belongs to is a child in a parent shipment group. The system determines the door by looking at the parent shipment.

You can also configure the loading criteria parameter "Accept Door Check Digit" if you want the system to force the user to enter the check digit for the dock door instead of the door barcode. After confirming the dock door, if there are no more required prompts, the system proceeds with loading updates for the LPN.

Non-Directed Loading (Load Trailer) Process

Identify Shipment

When entry mode is set to 'Shipment', 'BOL', or 'Dock Door', the system prompts for one of these values in the loading transaction (assuming Menu Mode) to identify a single shipment. This initiates a Non-Directed Loading mode in which the user can load any number of oLPNs onto the identified shipment. You can identify the shipment in several ways – either by the user directly providing the Shipment ID, or indirectly by the user providing a Bill of Lading number (which is tied to one of the pickup/delivery combinations on the Shipment) or a Dock Door ID (to which the shipment is already associated).

Accept Shipment Number

When entry mode is set to 'Shipment', the system initially prompts the user to enter a Shipment ID. The system runs several validations against this shipment.

- The shipment must be valid and present in the system.
- The shipment must have a pickup stop at the user's current warehouse. The system displays an error message if there is not a pickup stop for the current warehouse.
- The Pickup Stop for the current warehouse on the shipment must not have a Warehouse Stop Status of 'Loading Complete.' This ensures that warehouse users are not allowed to further load any product against a pickup stop which is already closed by the facility.
- The system also determines if the shipment is a parent shipment or if the shipment is part of a parent shipment group.
  - If it is, then the system identifies the parent shipment and allows you to load oLPNs belonging to any of the child shipments for the parent shipment.
  - Else if the shipment is not part of a parent shipment, then the system uses this shipment to validate against when the user attempts to load an oLPN.

Accept Bill of Lading Number

When entry mode is set to 'Bill of Lading', the system initially prompts the user to enter a Bill of Lading Number. The system generates this number at the time of shipment planning for each unique origin/Destination pair on a given shipment. The system attempts to identify the shipment id based on the bill of lading number. The system performs several validations for the BOL:

- The BOL number entered must match an existing barcode format for BOL Barcodes.
- The BOL number must be associated with an origin/destination pair in which the origin (pickup) is the current facility, and the origin pickup stop's warehouse status is not 'Loading Complete'.

Once the system performs all validations, it uses the derived shipment ID for the loading process.

Accept Dock Door

When entry mode is set to 'Dock Door,' the system initially prompts the user to enter a Dock Door. This input accepts either a dock door barcode, check digit, or dock door display location ID using the toggle/carousel on the mobile app.

> **Note**: Scanning the Location ID for the Dock Door is not supported.

Several validations occur at the time of Dock Door capture:

- Dock Door barcode (if entered) must match a valid barcode format with type = "dock door".
- Dock Door must be valid and present in the system.

Once the dock door is validated, the dock door is used to identify the shipment which is associated to the dock door.

- Loading checks if the dock door has a shipment assigned to it. If a shipment ID is assigned to the dock door, the system uses the shipment ID input for loading.
  - If a Parent Shipment ID has been assigned to the door, the oLPNs belonging to any of the Child Shipments can be loaded in the transaction.
- If there is no shipment assigned, the system also check for an existing reservation for the door, which is created as part of an in-progress yard move after yard check-in. If a reservation exists, the system uses the derived shipment ID as the shipment ID input for loading.
- If no assignment or reservation exists, Loading checks that the dock door is in an Available status (i.e. not in In-Use or Pending In-use). If door is available, the system prompts the ser to provide the shipment ID. See Accept Shipment Number section above as a reference for this input.

Accept oLPN

In non-directed mode, once you capture/identify a shipment ID, the system requests the user to scan an oLPN to load into the shipment. This is the same prompt that occurs after completing the loading of the previous oLPN inside the same shipment. However, in the non-directed mode, the validations are slightly different.
As per the directed mode, the system performs the following validations:

- The oLPN must be valid and present in the system. This validation checks that the scanned value is either an oLPN ID or an oLPN Cross-reference ID.
- The oLPN must be in a loadable status - at least packed and not already loaded, and not canceled.
- If the oLPN is planned on a shipment, and if stop sequence validation is enabled, then these validations occur at this point as well. See below for specific validations for enforcing correct delivery stop sequence loading.
- If the oLPN is planned on a shipment, and if loading sequence validation is enabled (stop sequence validation must be disabled), then the system will ensure that the oLPNs for the shipment are being loaded in an ascending loading sequence. See below for specific validations for enforcing correct loading sequence validation.
- If the oLPN is not on a shipment, then you can optionally perform shipment assignment (Fluid Loading) as a part of the non-directed load trailer flow. Please see the section below describing fluid loading.

Fluid Loading

When enabled, fluid loading allows the warehouse to dynamically assign oLPNs to an existing shipment if the oLPN's shipping attributes sufficiently match the shipment's same attributes. If the oLPN being loaded passes validation, the system assigns it to the shipment and adds a corresponding delivery stop to the shipment.

> **Note**: Fluid loading can only be used in non-directed mode. Fluid loading is **not** supported for Parent Shipment Groups.

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Common Features in the Loading Process

The following sections describe screens, features, and capabilities that are common to both directed and non-directed loading.

Trailer Capture

You can also optionally capture the trailer number as part of the loading process in both directed and non-directed modes, after the Shipment ID is captured. This allows the warehouse to stamp the trailer number on a shipment for purposes of transportation tracking. It is not a required entry as many flows may not know or need to track the trailer attribute.

When the Capture Trailer parameter is set to 'Do Not Accept Trailer Number', the system does not prompt for a trailer number in any loading process. If the Capture Trailer parameter is set to 'Accept Trailer Number", then after identifying the shipment, the system checks if the trailer has been captured already for that shipment (Trailer ID is populated on the shipment). If not, the system prompts the user to scan the trailer at that time.

The system performs the following validations:

- If set to 'Accept Trailer Number,' the barcode must be a valid barcode format with barcode type = "Trailer".
- Manhattan Active WM also validates that the trailer is a valid trailer that exists in the Yard Facility associated with the Warehouse Facility (this tie is made through the Warehouse Yard Mapping configuration screen in the UI). If the warehouse is linked to a Yard Facility and the trailer does not exist in the yard, then the system displays error DCS::265 is shown to the user - "Trailer is not Checked in. Proceed loading?" If you accept the error, the system allows the loading to continue.

Confirm Trailer Position

As a part of loading, the system can also be configured to display a trailer position confirmation screen as the last step in the process. This applies in both directed and non-directed modes.
When loading a pallet (physical pallet oLPN or pallet of oLPNs) and if the *Display Trailer Position When Loading Pallets* parameter is enabled, the solution checks the value selected in the *How do you want to confirm the trailer position?* parameter. See Loading and Unloading Configuration Guide.

- If this is set to *Tap to confirm planned position*, the solution follows the below logic:
  - If the trailer row/position is populated, then the system displays these values to the user. The user taps to confirm, and the system performs the loading updates.
  - Otherwise, if the pallet positions are not populated or NULL, the solution skips this screen and performs the loading updates.
- If this is set to *Display planned position, allow user to select any position*, the solution displays the planned position (or blank position if it is not planned) and prompts the user to enter a position. The user can directly enter a position or select a position from the drop-down. The drop-down displays all positions defined on the equipment layout. The solution does the following checks:
  - If the user enters or selects the same position, the solution loads the pallet and does not update the trailer row and position.
  - If the user selects a new position for a pallet, the solution validates the maximum UOM quantity at the location. If the UOM quantity is exceeded, the solution displays a warning message to the user, which can be overridden. Once the warning is overridden, the solution updates the trailer row ID and trailer position ID on the pallet and performs the loading updates.
  - If the user selects a NULL position, the solution displays an error message.

> **Note**: This can be configured as a warning and can be overridden. Once the warning is overridden, the solution updates the trailer row ID and trailer position ID on the pallet as NULL.

- If the user enters a position that does not exist on the equipment layout, the solution displays an error.

> **Note**: This can be configured as a warning and can be overridden. In this case, only the trailer position ID is updated with the value. The row is left blank.

Parent Shipment Assignment

Manhattan Active WM can optionally assign a parent shipment at the time of loading. This process is for supporting the LTL shipment flow where multiple shipments are grouped on a single trailer. When the “Enable Parent Shipment Assignment” parameter is turned on, the system validates the oLPN being loaded and the shipment at the door to see if it can be assigned.

- If the oLPN is currently assigned to a shipment (without a parent shipment) that has not begun loading, and the shipment resources (carrier/mode/service level) do not conflict with the parent shipment resources, then the system assigns the oLPN and shipment to the parent shipment.
- Else if the oLPN is currently assigned to a shipment (with a parent shipment) that has not begun loading, and the shipment resources (carrier/mode/service level) do not conflict with the parent shipment resources, then the system reassigns the oLPN and shipment to the new parent shipment.
- Else if the oLPN is currently assigned to a shipment (with a parent shipment) that has begun loading, the system will prevent the user from loading the oLPN since one or more oLPNs is already loaded and associated with a different parent shipment.

When the oLPN is successfully assigned to the parent shipment, then the solution also links the child shipments to the parent shipment and updates all other oLPNs on child shipments with a link to the parent shipment. This logic applies to individual oLPNs as well as oLPNs on a pallet. In the pallet flow, all oLPNs on the pallet must have a shipment and be eligible for assignment to the parent shipment. If one oLPN cannot be assigned to the parent shipment, the other oLPNs on the pallet are not assigned or loaded.

This flow is supported in various modes:

- Directed Loading “Load oLPN”. This is supported when the Determine Dock Door Determination parameter is set to:
  - Determine Dock Door By Shipment
  - Determine Dock Door by Consolidation Location
- Non-Directed Loading “Load Trailer”. This is supported for the following entry modes:
  - Shipment
  - Dock Door

Multi-Shipment Pallet Loading

Loading a pallet with multiple shipments is supported in the parent shipment flow. Since the parent shipment represents the logical grouping of shipments that can be shipped together, the solution allows simultaneously loading a pallet for a single parent shipment. This feature is supported for both Load oLPN and Load trailer flows.

Load oLPN Flow

Accept oLPN/Pallet

- The pallet must be valid and present.
- The oLPNs on the pallet must be in a valid status for loading – at least packed, not already loaded, and not canceled.
- The solution checks the shipment ID of the pallet to determine if it is a single shipment pallet.
  - If it is a single shipment pallet, it proceeds with standard oLPN validations for loading.

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- Otherwise, if multiple shipments on the pallet and one or more oLPNs on the pallet are not assigned to a shipment, the solution displays an error message to prevent loading this pallet.
- Also, if multiple shipments are on the pallet and all oLPNs have shipments, the solution proceeds to the next screen.

Dock Door Determination

The below logic describes how the solution determines the dock door.

- No determination of dock door: Proceed to load each oLPN to their respective shipments.
- Determine dock door by Shipment: Look at the parent shipment ID from the oLPNS on the pallet.
  - If a dock door is found on the parent shipment, the solution displays the dock door. When the user scans the dock door, the solution proceeds to load each oLPN to their respective shipments.
  - Else, it displays the error message, *DCS::328 "Dock door location cannot be determined for shipment*
    - If this message is set to warning and the user overrides the warning, the solution displays a dock door scan. If the user scans an eligible dock door that is available for assignment, the solution validates if there are multiple parent shipments on the current pallet.
      - If multiple parent shipments are on the pallet, the solution displays the error message, *DCS::391 Unable to assign dock door for scanned olpn {lpnId}. Multiple parent shipments present. Please assign 1 parent shipment to the door or repalletize*
      - Otherwise, the solution assigns the parent and child shipments to do the dock door and proceeds to load each oLPN for their respective shipments.

- Determination of dock door by Consolidation Location: Look at the current location of the pallet to determine if the consolidation location has an associated dock door.
  - If there is an associated dock door, then the solution displays the dock door. When the user scans the dock door, the solution proceeds to load each oLPN to their respective shipments.
  - Else, it displays the error message, *DCS::327 "Dock door location cannot be determined for shipment*

Load Trailer Flow

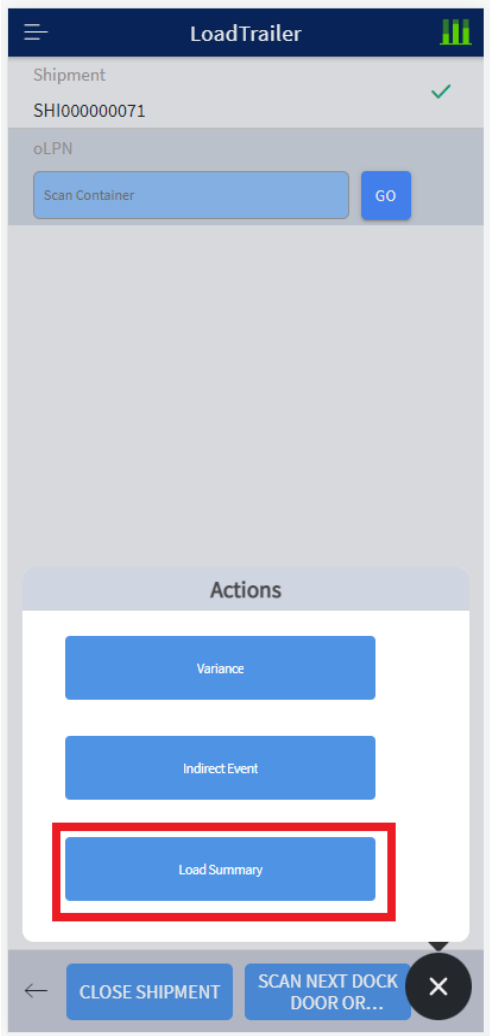
Parent shipment loading is supported for the load trailer flow when the entry mode is either Dock Door or Shipment (Parent Shipment). After identifying the parent shipment, the solution prompts for the oLPN/pallet. When the user scans a pallet, the solution validates the parent shipment on the oLPNs against the previously identified parent shipment.

- Suppose the parent shipments do not match or there is more than one parent shipment on the pallet. In that case, the solution can reassign the parent shipment based on configuration. See the Parent Shipment Reassignment section for more details.
- Otherwise, if the parent shipments match, the solution loads each OLPN to the individual child shipments.

> **Note**: When the "Allow Loading Multi-shipment Pallets to Parent Shipments" parameter is enabled, the solution, by default, removes the pallet ID from the loaded oLPNs.

Load Summary Screens in Mobile

The Mobile Loading screens are equipped with a load summary menu option for checking the progress of loading activity. To access the Load Summary, you must be in the context of a loading transaction such as Load oLPN or Load Trailer. From there, tap the pop-up menu in the lower right-hand corner of the app and select the Loading Summary option.



The Load Summary has header information for the shipment as well as running totals for weight and volume, which the system updates when you load the oLPNs. The summary tile widgets show counts and the percentage completed for stops as well as oLPNs.

Note: In order for Stop Completion on the Stop summary tile to update while loading is in progress, you must have the "Validate Stop Sequence" parameter enabled on the Loading Criteria as shown in the "Delivery Stop Sequence Validation" section above.

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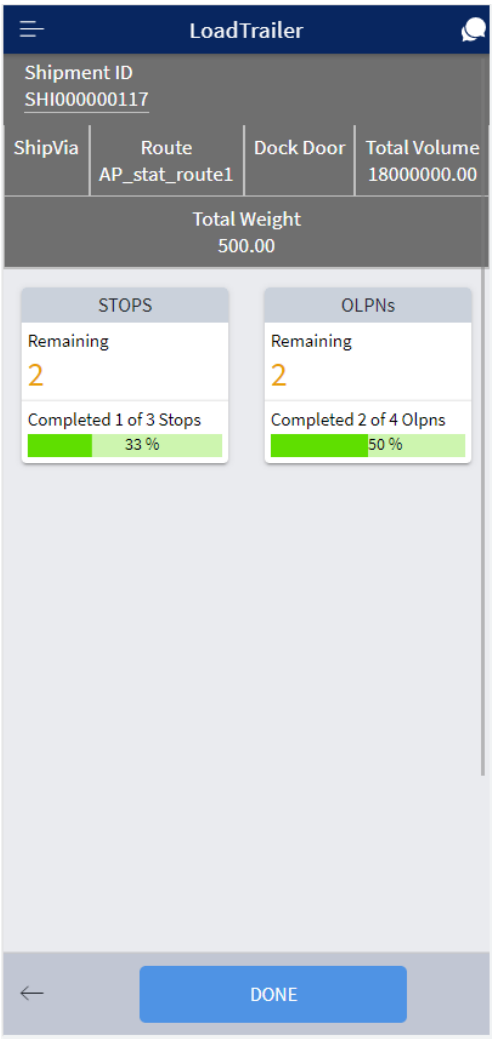
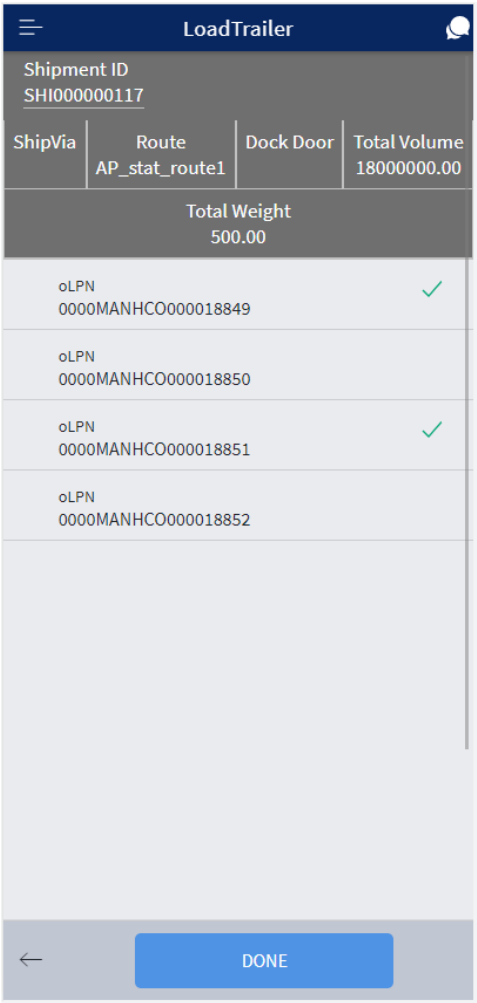
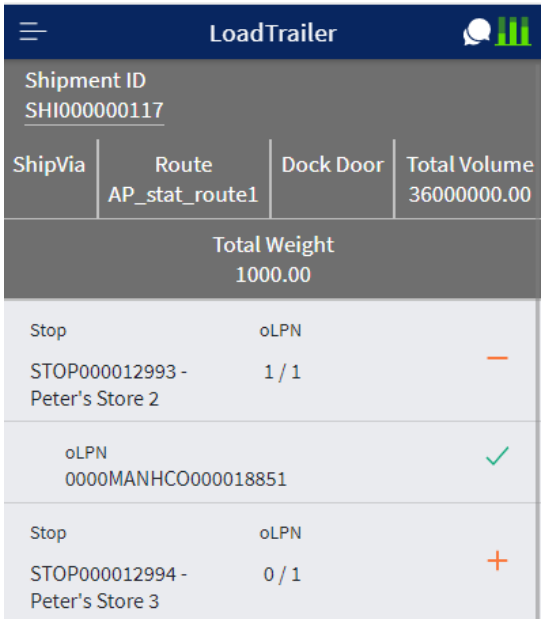


If you tap on one of the summary tiles, you can see more detailed information regarding the status of individual stops or oLPNs. This example shows the oLPN summary details.



Tapping on the Stop summary tile displays a list of the stops in order of how they are to be loaded on the truck and for each stop shows the count of oLPNs completed out of the total expected. You can tap on the "+" icon to expand stop details showing the specific oLPNs belonging to the stop. The check mark icon indicates the oLPNs that have been loaded.



Common FAQs for Load Summary

1. Why is my loading summary screen missing the "Current Stops" tile? Or why is the count of pallets incorrect under the oLPNs tile?
   Answer: These can be resolved by increasing the Shipping Parameter's oLPN Page Size to a higher value on the screen (the default is 20). While constructing the load summary screen data in the backend, MAWM does a lookup for oLPNs based on the oLPN page size.
2. Why is #the number of completed/remaining stops incorrect?
   Answer: MAWM will not conduct any stop level validation during loading and/or for the load summary screen if the Loading criteria "*Should this criteria validate Stop Sequence?*" for the corresponding loading transaction is set to False. This must be set to true for MAWM to validate against stops and also display the correct data in the load summary screen.

NMFC Freight Class Determination by Density

The solution calculates the National Motor Freight Classification (NMFC) freight class by density for NMFC regulations at an oLPN header level. Using this, users can standardize the evaluation of all commodities and classify them based on density as the primary characteristic.
Since this is an async process, data related to freight classification and code might take a few minutes to populate in the BOL report when loading is performed.
When the *Determine Freight Class Based on Freight Density* parameter is enabled, the solution first checks whether any item in the oLPN determines its freight class using density. The solution does the following:

- If no items use density, the solution assigns the highest freight class already on the items.
- If an item uses density, the solution does the following:
  - Calculates density (oLPN weight ÷ oLPN volume in lb/cubic feet) and rounds it down to two decimals.
  - Finds the matching NMFC class and then assigns the highest value between that density-based class and any fixed item freight classes.
  - Populates the Classification Code and NMFC Freight Class fields on the oLPN screen.
    - If all items share the same classification code, that code is assigned to the oLPN.
    - If items differ, the solution assigns the classification code of any item whose freight class matches the final oLPN freight class. If no match is found, the field is left blank.

Labor Tracking

The system tracks the loading against the configured labor activity.

- While loading, labor tracks the user's travel locations as well. This includes the reporting of oLPN's current location and the destination location (usually the dock door location where loading is complete).
- While loading/unloading, the system also has the provision to track the total units that are in the LPN or pallet. This allows customers to track and report activities at the unit level rather than just at the lpn level.

<u>Directed Loading:</u>

- When a user scans an oLPN and loads it at a directed dock door, dcshipping reports the oLPN's source location and sends data to create LM detail message with the oLPN information and dock door location.

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- LM creates a detail message with transaction sequence zero using the source location sent by WM. The loaded flag is set to 'Yes'. (This detail is not reported if the oLPN does not have any current location stamped)
- LM creates details message with oLPN information along with dock door location. The loaded flag is set to 'No' in this message.

Non-Directed Loading:

- When user scans the dock door, DC shipping sends dock door location only to create a LM detail message with dock door location only.
  - LM creates this detail message with location as dock door location. The loaded flag is set to 'No' in this message.
- For every oLPN scan (after the dock door scan), dcshipping sends data to create two LM detail messages:
  - LM detail message with oLPN information and oLPN's current location. The loaded flag is set to 'Yes' in this message. (This detail is not reported if oLPN does not have any current location.)
  - LM detail message with oLPN information and dock door location. The loaded flag is 'No' in this message.

Loading Task execution:

- When user executes the loading task, for every task detail execution, dcshipping sends data to create two LM detail messages:
  - LM detail message with oLPN information and oLPN's current location. The loaded flag is set to 'Yes' in this message. (This detail is not reported if the oLPN does not have any current location.)
  - LM detail message with oLPN information and dock door location. The loaded flag is 'No' in this message. (dock for location is not reported if the user is not confirming the dock door during loading task execution).

Actions

The system provides the user with a set of actions that they can perform while in the loading transaction. These actions are available either at the bottom of the screen or by expanding the action menu by clicking the arrow.

- Load Summary: This action displays the load summary screen. Please refer to the section above for details.
- Variance: This displays the list of oLPNs assigned to the shipment that are not loaded yet.

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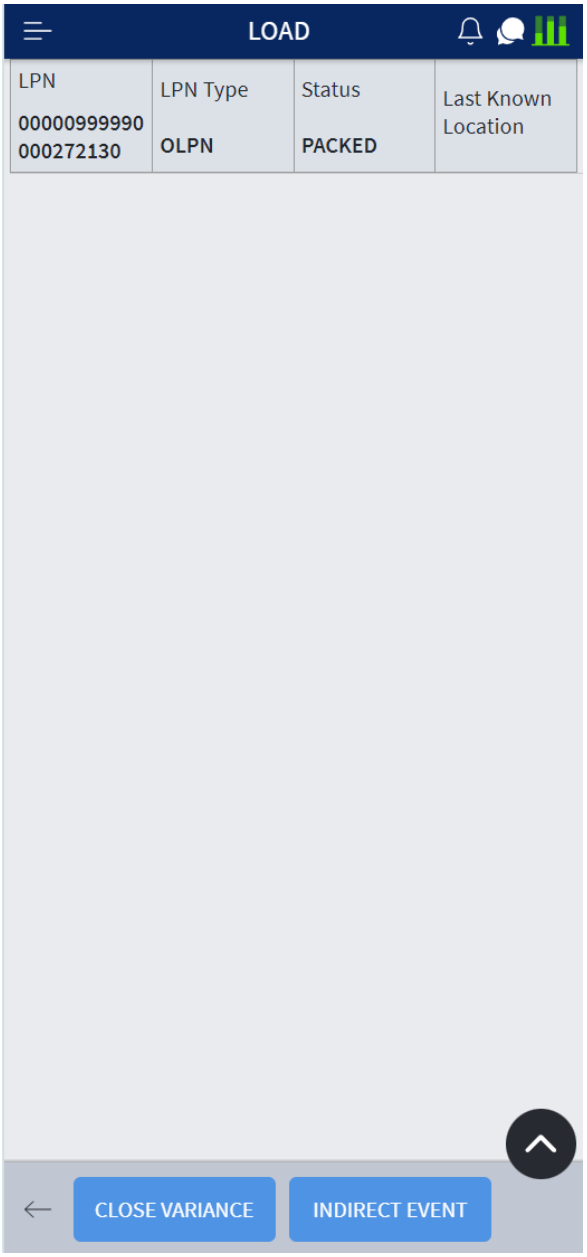
- 

- Asset Capture: You can capture an asset number through this action. Please refer to the Asset Tracking guide for more details.
- Indirect Event: When a user encounters some type of work that they must do that is not tracked as a labor activity in the system, they can use this action to record the activity. Please refer to the Indirect Events Process Guide for more details.
- Performance: A user can press this action to check how they are performing against a labor standard.

Validations

Delivery Stop Sequence Validation

TL Mode (Truckload) Shipments can be planned with multiple delivery stops. Trailers may leave the warehouse and traverse through a series of delivery stops. In these cases, the stops are sequenced in the order that they should be delivered on the route. The carrier typically requests the warehouse to load the product in reverse sequence of the delivery stop sequence, such that the first stop to be loaded in the nose of the trailer is the last delivery stop, and the last stop to be loaded is the first delivery stop. This 'Last In First Out' loading sequence ensures easy unloading at each delivery, since the product for the delivery is always immediately

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accessible in the trailer when delivered at the stop.

You can ensure that the product is loaded properly from nose to tail by using the loading transaction in Manhattan Active WM. This feature is enabled via a Loading Criteria parameter: "Validate Stop Sequence". When this feature is enabled, Manhattan Active WM validates the stop ID associated with each oLPN during loading compared to the other stops on the shipment. The system displays a warning message if the user attempts to load an oLPN that belongs to a subsequent stop when there are oLPNs for a higher stop sequence that are yet to be loaded.
Since Stop Sequence loading applies to TL Mode Shipments and not LTL mode, the Validate Stop Sequence parameter is **not applicable** to loading Parent Shipment groups. The model for parent shipments is that of a trailer full of LTL shipments with disparate destinations all being taken to the same sort hub, rather than a trailer being sent out on a route to make multiple deliveries. Therefore, the system does not treat child shipments as sequenced stops.

Loading Sequence Validation

Loading sequence is a fine-grained control mechanism that defines the exact order in which items (e.g., pallets or oLPNs) are loaded onto a trailer which optimizes physical placement of goods in the trailer to match unloading order and operational constraints. Loading Sequence and Stop Sequence do not always go hand-in-hand. Therefore, the system cannot validate both the stop sequence and the loading sequence for the same loading transaction/criteria. Users need to disable stop sequence in order to have the chance to enable loading sequence. Loading sequence is always validated in ascending order during loading.
Loading sequence validation in MAWM takes into account the lexicographical string comparison logic. Comparison always goes from left to right.
Example: So if there are two oLPNs with loading sequences 00004 and 01, "00004" comes before "01" as comparison goes from left to right and the first '0' == '0', but the next: 0’ (in “00004”) vs ‘1’ (in “01”), since '0' < '1' in ASCII order, "00004" comes before "01" lexicographically.
Loading sequence is always validated in ascending order. For users executing loading tasks, they have the ability to create loading tasks in ascending or descending order of loading sequence of the oLPNs. This can be done through the task sequencing and task break criteria within the loading task creation template in the loading task creation strategy. Even if the loading tasks are created in the descending order (say for operational reasons), the loading sequence validation during loading will always be carried out in the ascending order of oLPN loading sequence.
Lastly, the default level of message level is set to WARNING for Loading sequence validation. Meaning the users can accept the warning and continue loading. However, the maximum permissible message level is ERROR. So customers can set the message level to ERROR for DCS::411 to completely block their users from loading out of sequence if they wish to.

Batch master validation

Manhattan Active WM also validates oLPNs to ensure that you can catch items with a held batch at the point of loading. The system performs the below validations during the oLPN scan:

- If the oLPN contains single or multiple details of which none has a batch number, the system does not perform batch master validations.
- If oLPN contains single or multiple details and one or more details have batch numbers then
  - The system checks if the oLPN detail’s item is tracking batch numbers or not
    - If the item is not tracking batch numbers, the system does not perform batch master validation.
    - If the item is tracking batch numbers then:
      - The system checks 'Batch Master Validation' configuration for that item.
        - If the Batch Master Validation is DISABLED for that item, the system does not perform batch master validation.
        - If the Batch Master Validation is not DISABLED for that item, the system fetches the corresponding batch master information and validates whether or not the batch is held.

- If the batch is held, the system displays an error message and does not allow the user to load the oLPN.
- When the user scans a pallet for loading, the system executes the above validation for each oLPN present on the pallet. The whole pallet is not allowed to be loaded if any of the oLPNs has failed the batch master validation.

oLPN Condition Code Validation

The system validates the oLPN for condition codes when performing the loading transaction. However, the system provides the ability to bypass this validation for a condition code group if the loading criteria is configured with an oLPN override condition code group. You can only configure one condition code group in the criteria. In the case of pallet scan, the system validates all oLPNs in the pallet. If any of the oLPNs have a condition code which is not overridden in the criteria, the system prevents the entire pallet from processing further.
Examples of Condition Codes: Batch Expired, Quality held

Note: Unload oLPN transaction does not support condition code validation.

Note: All the loading errors and warnings for the given oLPN or the pallet (and the oLPNs on the pallet) are fetched simultaneously (during loading).
All errors are displayed (even if the errors are part of the same session). If there are no errors (and only warning or info messages), they are displayed continuously for acceptance or rejection during the pallet scan (for all the oLPNs).

Updates

Loading
During loading, after either Dock Door scan or oLPN scan if no Dock Door is confirmed, the system performs Loading updates. This involves the following updates:

1. Update oLPN to a Loaded status
2. Increment the Loaded Quantity/quantities on the Order line(s) with the quantity in the oLPN for each line.
3. Update the status of the Order line(s) for the oLPN to Loaded if Loaded Quantity = Order Quantity as a result of the loading quantity update.
4. Update the shipment's pickup stop warehouse status to Loading in Progress if this is the first oLPN to be loaded for the pickup stop
5. Mark the task detail as completed for the oLPN.

Unloading
Once the oLPN is successfully scanned in the unloading process, the system updates the oLPN to the "Packed" status.

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